

functable

Sign, variation and value tables for French mathematics

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1. Overview

functable provides two functions for drawing the standard French high-school mathematics tables:

- **sign-table** — tableau de signes et de variations (sign/variation/convexity tables in the *tkz-tab* style)
- **fun-table** — tableau de valeurs (function value table)

Both functions support auto-computation: provide a Typst closure and the values or signs are computed automatically.

2. sign-table

2.1. Basic usage

A sign table contains a header row (x), one or more **factor rows** whose signs multiply to give the **summary row**, an optional **variation row** with diagonal arrows, and optionally a second block for f' with a **convexity row**.

```
#sign-table(  
  factors: (  
    (expr:  $x - 1$ , zeros: ((value:  $1$ , approx:  $1$ ),), signs: ("-", "+")),  
    (expr:  $x + 2$ , zeros: ((value:  $-2$ , approx:  $-2$ ),), signs: ("-", "+")),  
  ),  
  summary-label:  $f'(x)$ ,  
  variation: true,  
  variation-label:  $f(x)$ ,  
)
```

x		$-\infty$	-2	1	$+\infty$	
facteurs	$x - 1$	$-$	$-$	0	$+$	
	$x + 2$	$-$	0	$+$	$+$	
$f'(x)$		$+$	0	$-$	0	$+$
$f(x)$						

2.2. Parameters

Parameter	Type	Default	Description
factors	array	()	Factor rows (see factor keys below).
signs	array	()	Direct signs for the summary row, without factor rows. Mutually exclusive with factors. Same values as a factor's signs.
zeros	array	()	Zeros attached to top-level signs / second-signs. Same

			format as a factor's zeros, including the number shorthand.
summary-label	content, none	none	Label for the f' summary row.
variation	bool	false	Show variation row with diagonal arrows.
variation-label	content, none	none	Label for the variation row.
variation-values	array	()	Values on the variation row: (at, label, pos).
bounds	auto, none, dict	auto	Domain bounds in x header. auto $\rightarrow -\infty/+\infty$. none hides both. Dict: (left: ..., right: ...).
start-value	content, none	none	Function value at the left edge of the variation row.
start-pos	string	"auto"	"top", "bottom", or "auto".
end-value	content, none	none	Function value at the right edge of the variation row.
end-pos	string	"auto"	"top", "bottom", or "auto".
col-width	length	1.5cm	Width of each interval column.
row-height	length	0.9cm	Height of sign/factor rows.
var-row-height	auto, length	auto	Height of variation and convexity rows. auto grows to fit the tallest label.
second-factors	array	()	Factor rows for f'. Same structure as factors.
second-signs	array	()	Direct signs for the f' block, without factor rows. Mutually exclusive with second-factors. Zeros come from top-level zeros.
second-summary-label	content, none	none	Label for the f' summary row.
convexity	bool, string	false	Convexity row. true/"symbol" $\rightarrow \cup/\cap$; "text" $\rightarrow$ "convexe"/"concave"; "both" $\rightarrow$ symbol above the word.
convexity-label	content, none	none	Label for the convexity row.
hd-fill	color	rgb("#cfe2f3")	Fill color when hd-style: "fill".
hd-style	string	"hatch"	HD band style: "hatch", "fill", or "blank".
show-facteurs	bool	true	Show rotated "facteur(s)" strip at the left spanning the factor rows.
background	color	white	Background colour for label knockout rectangles. Set to match your page or container fill.

2.2.1. Factor dictionary keys

Each entry in `factors` or `second-factors` is a dictionary. **Exactly one of `signs` or `fn` is required**; every other key is optional:

Parameter	Type	Default	Description
<code>expr</code>	content	—	<i>Optional.</i> Math expression for the row label (e.g. $x - 1$).
<code>zeros</code>	array	()	<i>Optional.</i> Zero declarations (see zero keys below).
<code>signs</code>	array	—	Required unless <code>fn</code> is given. Sign per interval: "+", "-", "" (blank), "HD". Takes precedence over <code>fn</code> .
<code>fn</code>	function	—	Required unless <code>signs</code> is given. $x \Rightarrow$ number for auto-sign inference. Return <code>none</code> to mark HD.
<code>interdit</code>	bool	false	<i>Optional.</i> All zeros of this factor are forbidden values of <code>f</code> (shorthand for <code>pole: true</code> on each zero).

2.2.2. Zero dictionary keys

Each entry in a `zeros` array is either a **plain number** — shorthand for (`value: $n$, approx: n`), convenient when the zero needs no special display — or a dictionary with the following keys:

Parameter	Type	Default	Description
<code>value</code>	content	—	Required. Displayed label (e.g. $\$1$, $\sqrt{2}$).
<code>approx</code>	float	—	Required. Numeric approximation, used for sorting, merging shared zeros, and test-point computation.
<code>pole</code>	bool	false	<i>Optional.</i> Valeur interdite: double bar in summary/variation/convexity rows; breaks arrows.
<code>mark</code>	content, "bar", "hd"	—	<i>Optional.</i> Custom marker in factor and summary rows. "bar" draws a double bar; "hd" continues the HD hatch through the point.
<code>summary-mark</code>	content, "bar", "hd"	—	<i>Optional.</i> Custom marker in the summary row only.

The number shorthand and the dictionary form can be mixed freely:

```
zeros: (-2, (value:  $\sqrt{2}$ , approx: calc.sqrt(2)), 3)
```

2.3. Direct signs — tables without factors

When the factorisation is not the point of the exercise, give the signs of the function directly with the top-level `signs` and `zeros` parameters instead of `factors` (the two are mutually exclusive):

```
#sign-table(  
  signs: ("+", "-", "+"),  
  zeros: (-2, 1),  
  summary-label:  $f(x)$ ,  
)
```

x	$-\infty$	-2	1	$+\infty$	
$f(x)$	$+$	0	$-$	0	$+$

The same mechanism gives a **variation table alone**, with no sign row at all — leave `summary-label` at none and the signs only steer the arrows:

```
#sign-table(
  signs: ("-", "+"),
  zeros: (2,),
  variation: true,
  variation-label: $f(x)$,
  start-value: $+oo$,
  end-value: $+oo$,
  variation-values: ((at: 2, label: $-3$)),
)
```

x	$-\infty$	2	$+\infty$
$f(x)$	$+\infty$	-3	$+\infty$

Per-zero options work as usual — mix the number shorthand with dictionaries to mark a pole:

```
signs: ("+", "-", "+"),
zeros: (-1, (value: $2$, approx: 2, pole: true)),
```

Symmetrically, `second-signs` drives the f'' block (and hence the convexity row) without `second-factors`; its zeros also come from the top-level `zeros` parameter. Note that a sign table **without** a variation table has always been the default — just leave `variation: false`.

2.4. Convexity display modes

The convexity parameter accepts, besides `false`, three display modes: `"symbol"` (equivalent to `true`) for the classic $\cup/\cap$, `"text"` for the words “convex”/“concave”, and `"both"` for the symbol stacked above the word:

```
#sign-table(
  second-signs: ("-", "+"),
  zeros: (0,),
  second-summary-label: $f''(x)$,
  convexity: "both",
  convexity-label: $f(x)$,
)
```

x	$-\infty$	0	$+\infty$
$f''(x)$	$-$	0	$+$
$f(x)$	$\cap$ concave		$\cup$ convexe

2.5. Variation values

Use variation-values to pin function values at interior zeros on the variation row:

```
variation-values: (
  (at: 2, label:  $-3$ ),           // auto-positioned
  (at: 5, label:  $0$ , pos: "top"), // forced top
)
```

pos accepts "top", "bottom", "arrow" (centred on the arrow line), or "auto".

2.6. HD (hors-domaine) intervals

Mark an interval as "HD" in signs (or return none from fn) for intervals where the expression does not exist. The interval renders as a hatched band:

```
#sign-table(
  factors: (
    (
      expr:  $1 / (2 \sqrt{x})$ ,
      zeros: ((value:  $0$ , approx:  $0$ , mark: "hd")),
      fn: x => if x <= 0 { none } else {  $1 / (2 * \text{calc}.\sqrt{x})$  },
    ),
  ),
  summary-label:  $f'(x)$ ,
  variation: true,
  variation-label:  $f(x)$ ,
  bounds: (left: none, right:  $+\infty$ ),
  start-value:  $0$ , start-pos: "bottom",
)
```

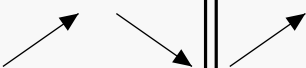
	x	0	$+\infty$
fact.	$\frac{1}{2\sqrt{x}}$		$+$
	$f'(x)$		$+$
	$f(x)$		

mark: "hd" continues the hatch **through** a boundary point — use this for a domain edge where f' is undefined but f continues (e.g. a vertical tangent), as opposed to mark: "bar" which draws an asymptote double bar.

2.7. Poles (valeurs interdites)

Set pole: true on a zero to draw a double bar $\parallel$ in the summary, variation and convexity rows, and to break the variation arrows at that point. Use the interdit: true shorthand on a factor when all its zeros are forbidden values of f :

```
#sign-table(
  factors: (
    (expr:  $x + 1$ , zeros: ((value:  $-1$ , approx:  $-1$ ),), signs: ("-", "+")),
    (
      expr:  $x - 2$ ,
      zeros: ((value:  $2$ , approx:  $2$ , pole: true),),
      signs: ("-", "+"),
    ),
  ),
  summary-label:  $f(x)$ ,
  variation: true,
  variation-label:  $f(x)$ ,
)
```

x		$-\infty$	-1	2	$+\infty$	
facteurs	$x + 1$	-	0	+	+	
	$x - 2$	-	-	0	+	
$f(x)$		+	0	-		+
$f(x)$						

2.8. Auto-sign inference with fn

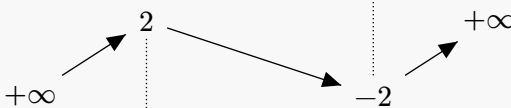
When signs is omitted and fn is provided, the sign in each interval is inferred by evaluating the function at a test point (midpoint of adjacent zeros for interior intervals, offset from the boundary zero for outer intervals):

```
//  $f(x) = x^3 - 3x$ :  $f'(x) = 3(x+1)(x-1)$ ,  $f''(x) = 6x$ 
#sign-table(
  factors: (
    (expr:  $x + 1$ , zeros: ((value:  $-1$ , approx:  $-1$ ),), fn:  $x \Rightarrow x + 1$ ),
    (expr:  $x - 1$ , zeros: ((value:  $1$ , approx:  $1$ ),), fn:  $x \Rightarrow x - 1$ ),
  ),
  summary-label:  $f'(x)$ ,
  variation: true,
  variation-label:  $f(x)$ ,
  start-value:  $+00$ ,
  end-value:  $+00$ ,
  variation-values: ((at:  $-1$ , label:  $2$ ), (at:  $1$ , label:  $-2$ )),
  second-factors: (
    (expr:  $6x$ , zeros: ((value:  $0$ , approx:  $0$ ),), fn:  $x \Rightarrow 6 * x$ ),
  ),
)
```

```

second-summary-label:  $f''(x)$ ,
convexity: true,
convexity-label:  $f(x)$ ,
)

```

x		$-\infty$	-1	0	1	$+\infty$	
facteurs	$x + 1$	$-$	0	$+$	$+$	$+$	
	$x - 1$	$-$	$-$	$-$	0	$+$	
$f'(x)$		$+$	0	$-$	$-$	0	$+$
$f(x)$							$+\infty$
fact.	$6x$	$-$	$-$	0	$+$	$+$	
$f''(x)$		$-$	$-$	0	$+$	$+$	
$f(x)$		$\cap$	$\cap$	$\cup$	$\cup$		

signs always takes precedence over fn if both are provided. Irrational zeros work correctly because test points are floats (e.g. for a zero at approx: `calc.sqrt(2)`, the test point in the right interval is `calc.sqrt(2) + 1.0`).

2.9. Simple table without factor strip

Set `show-facteurs: false` to hide the rotated “facteur(s)” label strip. Useful when there is only one factor row or when the factors are implied by context:

```

#sign-table(
  factors: (
    (zeros: (
      (value:  $-2$ , approx:  $-2$ ),
      (value:  $1$ , approx:  $1$ ),
    ), signs: ("+", "-", "+")),
  ),
  summary-label:  $f'(x)$ ,
  variation: true,
  variation-label:  $f(x)$ ,
  show-facteurs: false,
)

```

x	$-\infty$	-2	1	$+\infty$	
	$+$	0	$-$	0	$+$
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$					

2.10. Coloured background

Zero labels, bound labels, and variation values use a knockout rectangle to visually break the table grid lines. By default this rectangle is white. When the table sits on a non-white background — inside a coloured box or on a tinted page — set background to match so the knockout is invisible:

```
// Set once to apply to all tables in a document or scope:
#let sign-table = sign-table.with(background: luma(230))
```

All examples in this manual are rendered inside a light-grey #f8f8f8 block. The manual sets background globally at the top so the labels always look correct.

2.11. Domain boundary (range not in domain)

Use mark: "hd" on a zero that lies at a domain boundary (not an asymptote): the hatch continues **through** the zero point, and the variation arrow starts at that edge. The zero at the boundary already appears as a column header — do **not** also set the left bound to the same value (that would create a duplicate label). Use bounds: (left: none, ...) instead:

```
// f(x) = sqrt(x), defined on [0, +∞). f'(x) = 1/(2√x), undefined at 0.
#sign-table(
  factors: (
    (
      expr: $frac(1, 2 sqrt(x))$,
      zeros: ((value: $0$, approx: 0, mark: "hd")),
      fn: x => if x <= 0 { none } else { 1 / (2 * calc.sqrt(x)) },
    ),
  ),
  summary-label: $f'(x)$,
  variation: true,
  variation-label: $f(x)$,
  bounds: (left: none, right: $+∞$),
  start-value: $0$,
  start-pos: "bottom",
)
```

x		0	$+\infty$
fact.	$\frac{1}{2\sqrt{x}}$		+
$f'(x)$			+
$f(x)$		0	

A function undefined on an entire interval (e.g. $f(x) = \ln(x^2 - 4)$, undefined for $x \in]-2, 2[$) requires a domain check. Use fn for each factor and return none outside the domain. Zeros that fall inside the excluded zone (here $x=0$ for $2x$) should not be listed in zeros since they do not lie in the domain of f:

```
// f(x) = ln(x^2-4), domain: |x| > 2. f'(x) = 2x/(x^2-4).
// x=0 is NOT listed as a zero: it is outside the domain of f.
#sign-table(
  factors: (
    (
      expr: $2x$,
      zeros: (),
      fn: x => if x * x <= 4 { none } else { 2 * x },
    ),
    (
      expr: $x^2 - 4$,
      zeros: (
        (value: $-2$, approx: -2, pole: true),
        (value: $2$, approx: 2, pole: true),
      ),
      fn: x => if x * x <= 4 { none } else { x * x - 4 },
    ),
  ),
  summary-label: $f'(x)$,
  variation: true,
  variation-label: $f(x)$,
)
```

x		$-\infty$	-2	2	$+\infty$
facteurs	$2x$	-			+
	$x^2 - 4$	+	0	0	+
$f'(x)$		-			+
$f(x)$					

3. fun-table

3.1. Basic usage

```
#fun-table(  
  x-values: ($0$, $1$, $2$, $3$, $4$),  
  rows: (  
    (label: $f(x)$, values: ($1$, $3$, $5$, $7$, $9$)),  
  ),  
)
```

x	0	1	2	3	4
$f(x)$	1	3	5	7	9

3.2. Parameters

Parameter	Type	Default	Description
x-values	array	()	x column headers. Numbers: displayed in math mode and forwarded to fn. Dictionaries (display, value): custom display with numeric value. Content: display-only.
x-label	content	x	Label for the x row.
rows	array	()	Function rows (see row keys below).
col-width	length	1.5cm	Column width.
row-height	length	0.9cm	Row height.
label-width	auto, length	auto	Label column width. auto sizes to fit the widest label.
stroke	stroke	0.5pt	Table border and separator stroke.
decimals	auto, int	auto	Decimal places for computed values. auto trims trailing zeros.

3.2.1. Row dictionary keys

Parameter	Type	Default	Description
label	content	—	Row label.
values	array	—	Explicit content cells. Use none for blank. Takes precedence over fn for non-numeric x.
fn	function	—	$x \Rightarrow$ number used when x-values entries are numeric. Return none for a blank cell.
format	function, none	—	number $\Rightarrow$ content for custom rendering. Default: smart integer/decimal.

3.3. Auto-computed values

Provide numbers in x-values and a fn closure — values are computed and formatted automatically:

```
#fun-table(  
  x-values: (-3, -2, -1, 0, 1, 2, 3),
```

```

rows: (
  (label:  $f(x) = x^2 - 1$ , fn:  $x \Rightarrow x * x - 1$ ),
  (label:  $g(x) = 2x + 1$ , fn:  $x \Rightarrow 2 * x + 1$ ),
),
)

```

x	-3	-2	-1	0	1	2	3
$f(x) = x^2 - 1$	8	3	0	-1	0	3	8
$g(x) = 2x + 1$	-5	-3	-1	1	3	5	7

3.4. Custom display with dictionary x-values

For irrational x values or non-standard display, use (display: ..., value: ...) entries:

```

#fun-table(
  x-values: (
    (display:  $-\pi/2$ , value: -calc.pi / 2),
    (display:  $0$ , value: 0.0),
    (display:  $\pi/2$ , value: calc.pi / 2),
    (display:  $\pi$ , value: calc.pi),
  ),
  rows: (
    (label:  $\sin(x)$ , fn:  $x \Rightarrow \text{calc.sin}(x)$ ,
      format:  $x \Rightarrow \text{\#str}(\text{calc.round}(x, \text{digits: } 4))$ ),
    (label:  $\cos(x)$ , fn:  $x \Rightarrow \text{calc.cos}(x)$ ,
      format:  $x \Rightarrow \text{\#str}(\text{calc.round}(x, \text{digits: } 4))$ ),
  ),
)

```

x	$-\pi/2$	0	$\pi/2$	π
$\sin(x)$	-1	0	1	0
$\cos(x)$	0	1	0	-1